

Faster Gaussian resampling and a conditional integer-multiplication saving above 2^{-30}

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Abstract

After the compressed complex network, the conditional witness $\kappa = 59/10^{11}$ was capped by the Gaussian resampling maps, whose cost $O(tp^{3/2+\delta}\alpha)$ per line forces the dimension exponent $\epsilon < 1/5$. We make two changes to those maps. First, a chirp identity writes every Gaussian sum along a line as one correlation per block, which the established integer multiplier evaluates in $O(p^{1+\delta})$ per output, independently of the Gaussian width. Second, a potential-function argument shows that the powers of the correction $E = N - I$ decay like $e^{-\pi\alpha^2\sigma n}$ after a burn-in of order α^2/θ bits, so the Neumann series needs $O(p/\alpha^2 + 1/\theta)$ terms rather than $p/(\alpha^2\theta)$. With $\alpha = \lfloor \sqrt{b/(8d)} \rfloor$ the Gaussian cost per bit becomes $O(d^2p^\delta)$, and ϵ may approach $1/2$. The compressed complex network leaves room for the larger guard parameter this needs. The result is the conditional witness $\kappa = 1479/10^{12} > 2^{-30}$, a factor $1479/590 \approx 2.51$ above the preceding witness. The underlying manuscript's interfaces remain assumptions.

1 Scope and dependencies

The source is OpenAI's *Integer multiplication below $n \log n$* , pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a,

with the compact-control construction and the compressed complex network. The computational model is a fixed finite alphabet and a fixed number of one-dimensional tapes. We retain the paired bit network ($\tau = 1 - 296/10^{11}$), the compressed complex network ($\sigma = 1 - 14/10^9$), the compact-control movement, reservation, repair and generalized guard, and all transform, error and rounding interfaces. Only the one-dimensional resampling algorithms, the Gaussian width α and the assembly parameters change. Numbered results of Harvey and van der Hoeven [1] refer to their author-hosted manuscript.

2 The one-dimensional interface

For integers $2 \leq s < t < 2^p$ with $\gcd(s, t) = 1$ and $2 \leq \alpha < \sqrt{p}$, the resampling argument of [1, Section 4] uses the Gaussian maps $S, T : \mathbb{C}^s \rightarrow \mathbb{C}^t$,

$$(Su)_k = \frac{1}{\alpha} \sum_{j \in \mathbb{Z}} e^{-\pi\alpha^{-2}(j-sk/t)^2} u_j, \quad (Tu)_k = \sum_{j \in \mathbb{Z}} e^{-\pi\alpha^2(tj/s-k)^2} u_j,$$

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the nearest indices $q_j = \lfloor tj/s + 1/2 \rfloor$ with $\beta_j = tj/s - q_j$, the selection $(Cv)_j = v_{q_j}$, the diagonal $(Du)_j = e^{\pi\alpha^2\beta_j^2}u_j$, and $N = \text{CTD}$. Put $\sigma = t/s$ and $\theta = \sigma - 1$. The cited Theorem 4.2 and Lemma 4.6 give

$$\mathsf{T}P_sF_s = P_tF_tS, \quad \|N - I\| < 2.01e^{-\pi\alpha^2\theta/2} < 2^{-\alpha^2\theta} \quad (\alpha^2\theta \geq 1). \quad (1)$$

The manuscript's no-sort interface keeps the permutations P_s, P_t in the transform identity and approximates $S' = S/2$, $J' = N^{-1}/2$ and $D' = 2^{2-2\alpha^2}D$. We change its hypotheses and time bounds as follows.

Lemma 1 (One-dimensional interface, revised). *Assume in addition*

$$\alpha^{-2} \leq \theta \leq \frac{1}{4}, \quad n_{\alpha,\theta} := \min\left\{\left\lceil \frac{p}{\alpha^2\theta} \right\rceil, \left\lceil \frac{p+1}{4\alpha^2} + \frac{1}{\theta} \right\rceil\right\}.$$

Then $P_sF_s = 2^{2\alpha^2}B_0P_tF_tA$ with contractions $A = S'$ and $B_0 = D'J'C$, and there are approximations $\tilde{A}$ and $\tilde{B}_0$ with errors less than p^2 , in time $O((t+p)p^{1+\delta})$ and $O(n_{\alpha,\theta}(t+p)p^{1+\delta})$.

Proof. The exact identity and the norm bounds $\|S'\| < 3/4$, $\|J'\| < 7/8$, $\|D'\| < 1$ are those of the manuscript: the cited Lemma 4.6 needs only $\alpha^2\theta \geq 1$, which gives $\|N - I\| < 0.42$, and the proof of the cited Proposition 4.7(i) uses only that bound, $\alpha \geq 2$ and the cited Lemma 4.5. The original hypothesis $\theta > p/\alpha^4$ was used only to bound the cost of the Neumann evaluation. Lemma 4.8 of [1] gives $\tilde{D}'$ with error less than 4 in time $O(tp^{1+\delta})$; Lemma 4 below gives $\tilde{S}'$ and $\tilde{E}$; and the Neumann evaluation of the cited Lemma 4.12, run for $n_{\alpha,\theta}$ terms with this $\tilde{E}$, gives $\tilde{J}'$ with error less than $3p^2/4$. Its analysis uses only $\|E\| < 1/2$, $\varepsilon(\tilde{E}) < p/3$, $n \leq p$ and $\|E^n\| \leq 2^{-p}$, with $\|E^{n+k}\| \leq \|E^n\|\|E\|^k$ for the tail; Corollary 3 supplies the last two. The selection C and the record adapters cost $O(tp)$ as before. $\square$

2.1 Faster Gaussian maps

This subsection supplies the two ingredients used in the proof of Lemma 1. We keep the notation of that proof and put $\sigma = t/s$, $\theta = \sigma - 1$. Define q_j and β_j by the same formulas for every integer j , so that $q_{j+s} = q_j + t$ and $\beta_{j+s} = \beta_j$. Write $E = N - I$. By [1, Section 4.2],

$$(Eu)_\ell = \sum_{h \neq 0} e^{-\pi\alpha^2 X_{\ell,h}} u_{\ell+h}, \quad X_{\ell,h} = (\sigma h + \beta_\ell)^2 - \beta_{\ell+h}^2,$$

with indices of u taken modulo s , and every $X_{\ell,h}$ is nonnegative.

Lemma 2 (Powers of the correction). *Assume $\alpha \geq 2$ and $0 < \theta < 1$. For every integer $n \geq 1$,*

$$\|E^n\| \leq e^{\pi\alpha^2(1/(4\theta)+1/2)} (2.01 e^{-\pi\alpha^2\sigma})^n.$$

Proof. Expanding the product, $(E^n u)_\ell$ is the sum over $h_1, \dots, h_n \in \mathbb{Z} \setminus \{0\}$ of $e^{-\pi\alpha^2 X_{\ell,h_n}} u_{j_n}$, where $j_0 = \ell$, $j_i = j_{i-1} + h_i$ and $X = \sum_i X_{j_{i-1}, h_i}$. Moving each subtracted square to the following step,

$$X = \sum_{i=1}^n c(j_{i-1}, h_i) + \beta_{j_0}^2 - \beta_{j_n}^2, \quad c(j, h) = (\sigma h + \beta_j)^2 - \beta_j^2 = \sigma h(\sigma h + 2\beta_j).$$

Put $\Phi(j) = \sigma(\beta_j^2 + 1/4)/\theta$. For one step (j, h) let $y = \beta_j + h\theta$ and $w = q_{j+h} - q_j - h \in \mathbb{Z}$. Then $\beta_{j+h} = \beta_j + \sigma h - (q_{j+h} - q_j) = y - w$, so $|y - w| \leq 1/2$. Substitute $\beta_j = y - h\theta$ in $c(j, h)$ and in $\Phi(j+h) - \Phi(j) = \sigma((y-w)^2 - (y-h\theta)^2)/\theta$. Expanding and using $\sigma - \theta = 1$ gives

$$c(j, h) - \Phi(j+h) + \Phi(j) = \sigma h^2 + \frac{\sigma}{\theta} w(2y-w) \geq \sigma h^2,$$

because $w(2y-w) = w^2 + 2w(y-w) \geq w^2 - |w| \geq 0$. Sum over the path. Since $\sigma/(4\theta) \leq \Phi \leq \sigma/(2\theta)$ and $\beta_{j_n}^2 \leq 1/4$,

$$X \geq \sigma \sum_i h_i^2 - \frac{\sigma}{4\theta} - \frac{1}{4} = \sigma \sum_i h_i^2 - \frac{1}{4\theta} - \frac{1}{2}.$$

Consequently

$$\|E^n\| \leq e^{\pi\alpha^2(1/(4\theta)+1/2)} \left(\sum_{h \neq 0} e^{-\pi\alpha^2\sigma h^2} \right)^n.$$

As $\pi\alpha^2\sigma > 4\pi$, the inner sum is less than $2e^{-\pi\alpha^2\sigma}(1 + 2e^{-3\pi\alpha^2\sigma}) < 2.01e^{-\pi\alpha^2\sigma}$. $\square$

The original estimate $\|E\|^n < 2^{-\alpha^2\theta n}$ gains only $\alpha^2\theta$ bits per term. The lemma gains about $\pi\alpha^2 \log_2 e$ bits per term after a burn-in of order α^2/θ bits.

Corollary 3. *Assume $\alpha \geq 2$, $\alpha^2\theta \geq 1$ and $\theta < 1$, and let $n_{\alpha,\theta}$ be as in Lemma 1. Then $n_{\alpha,\theta} \leq p$ and $\|E^{n_{\alpha,\theta}}\| \leq 2^{-p}$.*

Proof. If $n_{\alpha,\theta} = \lceil p/(\alpha^2\theta) \rceil$, then $n_{\alpha,\theta} \leq p$ and $\|E^n\| \leq \|E\|^n < 2^{-\alpha^2\theta n} \leq 2^{-p}$ by (1). Otherwise $n = n_{\alpha,\theta} \geq (p+1)/(4\alpha^2) + 1/\theta$ and $n < \lceil p/(\alpha^2\theta) \rceil \leq p$. Put $\kappa_0 = \pi \log_2 e$, so $4.531 < \kappa_0 < 4.534$, and note $\log_2 2.01 < 1.01$. Taking base-two logarithms in Lemma 2,

$$\log_2 \|E^n\| \leq \kappa_0 \alpha^2 \left(\frac{1}{4\theta} + \frac{1}{2} \right) - n(\kappa_0 \alpha^2 \sigma - 1.01).$$

Since $\sigma > 1$ and $\alpha \geq 2$, the last bracket is at least $4\alpha^2$. Also $4.534(1/(4\theta) + 1/2) \leq 4/\theta$ because $\theta < 1$. Hence $\log_2 \|E^n\| \leq 4\alpha^2/\theta - 4\alpha^2 n \leq -p - 1$. $\square$

Lemma 4 (Gaussian maps by chirped correlation). *Assume the hypotheses of Lemma 1. There are fixed multitape algorithms computing approximations $\tilde{S}' : \mathbb{D}_p^s \rightarrow \mathbb{D}_p^t$ of S' and $\tilde{E} : \mathbb{D}_p^s \rightarrow \mathbb{D}_p^s$ of E , with errors less than $16p$ and $p/3$ respectively, outputs in the disk grid, and time $O((t+p)p^{1+\delta})$ each. Inputs and outputs occur in increasing coordinate order. The constants are uniform in p, s, t, α .*

Proof. Chirp identity. For a rational y and integers a, b , put $\hat{a} = a + y/\sigma$. Expanding both sides and using $\sigma - \theta = 1$,

$$(\sigma\hat{a} - b)^2 = \sigma\theta\hat{a}^2 + \sigma(\hat{a} - b)^2 - \theta b^2.$$

Hence, for rational $\lambda > 0$,

$$e^{-\pi\lambda(\sigma a + y - b)^2} = e^{\pi\lambda\theta b^2} e^{-\pi\lambda\sigma\theta(a+y/\sigma)^2} e^{-\pi\lambda\sigma(a-b+y/\sigma)^2}. \quad (2)$$

For inputs f_a , $0 \leq a < L_A$, the sums $F_b = \sum_a e^{-\pi\lambda(\sigma a + y - b)^2} f_a$, $0 \leq b < L_B$, are therefore $F_b = e^{\pi\lambda\theta b^2} c_b$, where

$$c_b = \sum_a g_a K_{a-b}, \quad g_a = e^{-\pi\lambda\sigma\theta(a+y/\sigma)^2} f_a, \quad K_r = e^{-\pi\lambda\sigma(r+y/\sigma)^2}.$$

This is one correlation of finite sequences; $K_r \in (0, 1]$ and $|g_a| \leq |f_a|$.

One block. Let $P \geq p$ be an integer, $|f_a| \leq F$ with $F \geq 1$, and $e^{\pi\lambda\theta b^2} \leq 2^\Gamma$ for $0 \leq b < L_B$, with $\Gamma \leq 2P$. Suppose approximations $\tilde{f}_a$ with $|\tilde{f}_a - f_a| \leq 2^{4-P}F$ are available. All exponents below are π times rationals whose numerators and denominators have $O(p)$ bits. Lemmas 2.13 and 2.14 of [1], at precision P , give the decaying factors and kernel values, and the numbers $2^{-\Gamma}e^{\pi\lambda\theta b^2}$, each with absolute error at most 2^{1-P} . Multiply each $\tilde{f}_a$ by its factor and round to precision P . Represent every resulting real or imaginary part as an integer multiple of 2^{-P} , split it into its nonnegative

and negative parts, and pack each sequence into one integer with slots of $2P + \lceil \log_2(FL_A) \rceil + 8$ bits. A constant number of products by the established multiplier [1, Theorem 1.1] then gives every rounded c_b exactly, because no slot overflows. Multiply by the approximate factor $2^{-\Gamma} e^{\pi\lambda\theta b^2}$, shift by Γ , and round to precision p .

The rounded g_a and K_r have errors at most $2^{5-P}F$ and 2^{1-P} , so the rounded correlation differs from c_b by at most $2^{7-P}FL_A$, while $|c_b| \leq FL_A$. After the shift, the final factor has absolute error at most $2^{\Gamma+1-P}$. Therefore the computed value differs from F_b by at most $2^{\Gamma+9-P}FL_A$, which is at most 2^{-p-2} whenever

$$P \geq p + \Gamma + \lceil \log_2(FL_A) \rceil + 11.$$

The block uses $O(L_A + L_B)$ exponentials and products at precision P and integer products on $O((L_A + L_B)(P + \log(FL_A)))$ bits. When $P = O(p)$ and $\log(FL_A) = O(p)$, its cost is $O((L_A + L_B)p^{1+\delta})$.

The map S' . Here $(S'u)_k = (2\alpha)^{-1} \sum_j e^{-\pi\lambda(\sigma j - k)^2} u_j$ with $\lambda = (\sigma\alpha)^{-2}$. Let $m = \lceil \sqrt{p} \rceil \alpha$, the window of [1, Lemma 4.9]. Partition $0 \leq k < t$ into consecutive blocks of at most m outputs. For a block starting at k_0 , take inputs $j = j_0 + a$, $0 \leq a < L_A = 3m + 3$, where $j_0 = \lfloor k_0/\sigma \rfloor - m$, with the periodic indices of u , and put $k = k_0 + b$, $y = \sigma j_0 - k_0$. These inputs contain every j with $|j - k/\sigma| < m$ for every k in the block. The block sums therefore omit only terms omitted by the truncated sums of that lemma, so they differ from $(S'u)_k$ by less than 2^{-p} . Here $f_a = u_{j_0+a}$ is exact, $F = 1$, and for $b < m$, $\pi\lambda\theta b^2 \leq \pi\theta(\sqrt{p} + 1)^2 \leq (\pi/4)(1.21p)$ because $\theta \leq 1/4$ and $p > 100$, so $\Gamma = \lceil 1.4p \rceil$ suffices. Take $P = 3p$, and apply the factor $(2\alpha)^{-1}$ by one more product at precision P . Each output then has error less than $2^{-p}(1 + 1/4 + 1) < 3 \cdot 2^{-p}$, and its norm is less than $\|S'\| + 3 \cdot 2^{-p} < 1$.

The map E . With $f_j = e^{\pi\alpha^2\beta_j^2} u_j$ and $\lambda = \alpha^2$, the definition of N gives

$$(Nu)_\ell = \sum_j e^{-\pi\lambda(\sigma j - q_\ell)^2} f_j,$$

and its term $j = \ell$ is exactly u_ℓ . The inputs f_j are formed by Lemma 2.14 of [1] for $2^{-\lceil 1.14\alpha^2 \rceil} e^{\pi\alpha^2\beta_j^2}$, a shift, and one product with u_j ; their errors are at most $2^{4-P}F$. Let $m = \lceil \sqrt{p}/(2\alpha) \rceil$, the window of [1, Lemma 4.11]. Partition $0 \leq \ell < s$ into blocks of at most m indices. For a block starting at ℓ_0 , take inputs $j = j_0 + a$, $0 \leq a < L_A = 3m + 1$, with $j_0 = \ell_0 - m$, and evaluate the correlation at every integer $b = q_{\ell_0} + b'$, $0 \leq b' < L_B = 2m + 2$; these include all q_ℓ of the block because $\sigma < 2$. Put $y = \sigma j_0 - q_{\ell_0}$. Then $F = e^{\pi\alpha^2/4} < 2^{1.14\alpha^2} \leq 2^{1.14p}$ and

$$\pi\lambda\theta b'^2 \leq \pi\theta\alpha^2(2m + 2)^2 \leq \frac{\pi}{4}(\sqrt{p} + 4\alpha)^2 < \frac{25\pi}{4}p,$$

so $\Gamma = \lceil 29p \rceil$ and $P = 34p$ suffice, using $(\sqrt{p} + 4\alpha)^2 < 25p$ and $\theta \leq 1/4$. Select the outputs $b = q_\ell$ in order, subtract the exact u_ℓ , and round. The block contains every term $0 < |h| \leq m$ of the truncated sum in that lemma, so the computed $(\tilde{E}u)_\ell$ differs from $(Eu)_\ell$ by less than $(3+1+1)2^{-p} < (p/3)2^{-p}$. Its norm is less than $\|E\| + 5 \cdot 2^{-p} < 1$.

Cost and tape movement. The selected indices q_ℓ , the numerators $s\beta_\ell$, and the block offsets are generated by the counters already used for C , in $O(p)$ time per index. Before a line, copy u with $3m + 3$ periodic records appended at each end, in $O((s + m)p)$ time; then every block reads one consecutive interval. Consecutive input intervals overlap in $O(m)$ records, so all reads, packing and unpacking cost $O((t + m)p)$. There are $O(t/m + 1)$ blocks, each costing $O(mp^{1+\delta})$; in both cases $m < 2p$. This gives the time bound $O((t + p)p^{1+\delta})$. $\square$

3 Parameters and assembly

The tensor interface applies the one-dimensional maps line by line. With $n_* = \max_i n_{\alpha, \theta_i}$ and every $t_i \geq p$, its time becomes $\sum_i (T/t_i) O(n_*(t_i + p)p^{1+\delta}) = O(dTp^{1+\delta}(1 + n_*))$.

Gaussian width. With $b = \lceil \log_2 n \rceil$, $p = 6b$, $d = \lfloor b^\epsilon \rfloor$ and $\eta = 1/(4d)$ as in the manuscript, set

$$\alpha = \left\lfloor \sqrt{b/(8d)} \right\rfloor, \quad \gamma = 2d\alpha^2 \leq b/4.$$

The bound $\gamma \leq b/4$ is the one used by the retained final rounding argument. For large n , $b \geq 96d$, so $\alpha \geq 2$ and $\alpha^2 \geq b/(16d)$; also $\alpha^2 < p$. The chosen primes satisfy $\eta < \theta_i < 1/(2d-1) \leq 1/4$, so $\alpha^2 \theta_i > b/(64d^2) \geq 1$ once $b \geq 64d^2$, which holds eventually because $2\epsilon < 1$. Hence

$$n_{\alpha, \theta_i} \leq \frac{p+1}{4\alpha^2} + \frac{1}{\theta_i} + 1 \leq \frac{(6b+1)16d}{4b} + 4d + 1 \leq 30d.$$

The original choice $\alpha = \lceil (32db)^{1/4} \rceil$ would need $\lceil p/(\alpha^2 \theta_i) \rceil$ Neumann terms, of order $(dp)^{1/2}$; with the larger width the old windowed Gaussian sums would cost $\Theta(p/\sqrt{d})$ times more per output than the chirped correlations.

Cost row. Each axis has at most T/t_i lines of length $t_i > p$, so the Gaussian maps cost $O(d^2 Tp^{1+\delta})$, that is $O(d^2 p^\delta)$ per bit, with power $2\epsilon + \delta$. This replaces the power $3/4 + \delta + 5\epsilon/4$. All other rows of the cost table are unchanged.

Parameters. Put $a = 296/10^{11}$, $a_c = 14/10^9$ and choose

$$\begin{aligned} \tau = 1 - a, \quad \sigma = 1 - a_c, \quad \epsilon = \frac{49999}{100000}, \quad c = \frac{9999}{10000}, \quad \beta = \frac{19}{25}, \quad \zeta = \frac{1}{10000}, \quad \delta = \frac{1}{10^6}, \\ C_1 = 5 - 4\beta + \zeta = \frac{19601}{10000}, \quad \lambda = 1 - \frac{29595}{10^{13}}, \quad \lambda' = 1 - \frac{2959}{10^{12}}, \quad \kappa = \frac{1479}{10^{12}}. \end{aligned}$$

Since $\sigma < \tau$, the compact-control internal exponent is $\chi = \tau$. Exact comparisons give

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c\} < \lambda', \quad \epsilon C_1 = \frac{980030399}{10^9} < 1.$$

At this ϵ the guard condition $\epsilon C_1 < 1$ forces $\beta > 3/4$, and the leaf condition forces $(1 - \beta)a_c > 1 - \lambda'$. Both hold only because the compressed complex saving a_c exceeds a by a factor 4.7; with the original complex saving $418/10^{12}$ no β satisfies both. The remaining setup conditions $\epsilon(1 + c) < 1$, $1 - 2\epsilon > 0$ and $\epsilon + \delta < 1$ hold.

The seven assembly margins are

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), \\ g_3 &= \epsilon(1 - \lambda'), & g_4 &= (1 - \tau)(1 - \epsilon), \\ g_5 &= 1 - \delta - 2\epsilon, & g_6 &= 1 - \delta - \epsilon, \quad g_7 = \epsilon, \end{aligned}$$

and exact arithmetic gives

$$G_* = \min_i g_i = g_3 = \frac{147947041}{10^{17}}, \quad G_* - \kappa = \frac{47041}{10^{17}} > 0.$$

This strict gap absorbs all remaining fixed powers of $\log p$, and the retained assembly yields

$$T(n) = O(n(\log n)^{1-1479/10^{12}}), \quad 2^{-30} < \frac{1479}{10^{12}} < 2^{-29},$$

conditional on the identified upstream and extension proofs.

The next ceiling. Now $g_2, g_3, g_4 \leq a \min\{\epsilon, 1 - \epsilon\}$, so these networks and assembly inequalities give $\kappa < a/2 < 2^{-29}$. The witness is above 99.9% of this bound. Further progress needs a stronger bit network; the complex network would then need to keep $(1 - \beta)a_c > a$ with $\beta > 3/4$.

4 Reproducibility and review boundary

`scripts/fast_gaussian.py` checks the witness, the guard, the exact step inequality of Lemma 2 on several (s, t) , and the Neumann counts; its certificate is `certificates/fast-gaussian.json`. `tests/test_fast_gaussian.py` verifies the chirp identity exactly, the power bound against explicit E matrices at 160 decimal digits, the block correlation against direct sums, negative parameter cases, and the patch text. `scripts/make_fast_gaussian_patch.py` writes `patches/fast-gaussian-30.patch`. These checks support the written arguments; they are not a formal verification of the upstream theorem, and the new construction has not received independent review.

References

- [1] D. Harvey and J. van der Hoeven, *Integer multiplication in time $O(n \log n)$* , Ann. of Math. (2) **193** (2021), no. 2, 563–617.